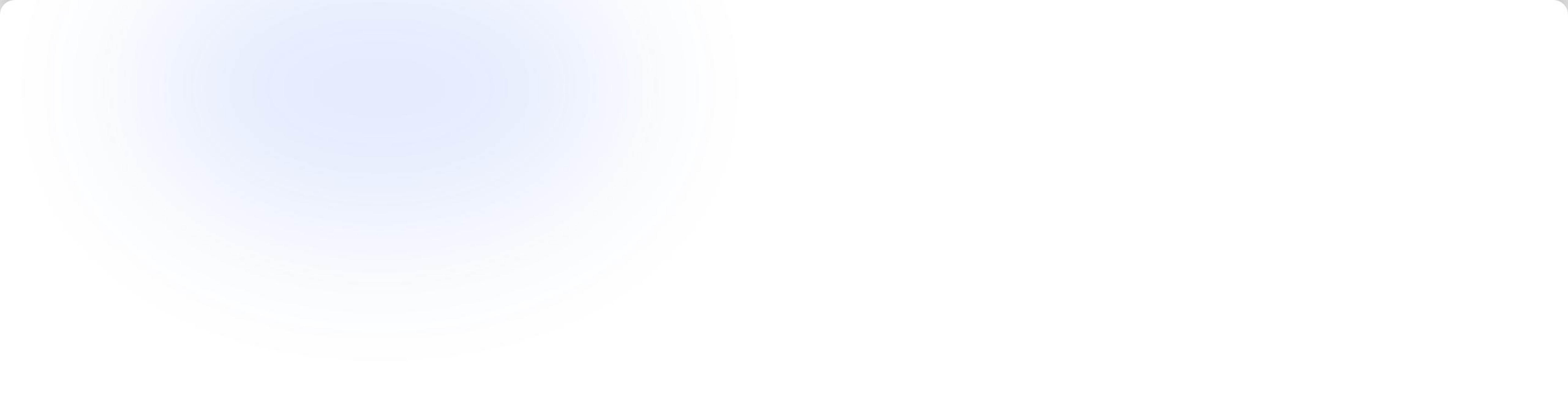


SpaceX Falcon 9 Landing Prediction

Machine Learning & Data Science Capstone Project | Predicting First Stage
Booster Recovery Success & Commercial Cost Advantage



IBM Data Science Professional Certificate



Executive Summary & Pipeline

An end-to-end data science methodology applied to rocket launch telemetry, web scraping, geospatial analytics, and predictive classification algorithms.

| Data Collection & Wrangling

- ✓ **SpaceX REST API (v4):** Queried launch records to extract payload mass, core serials, orbit types, and landing outcomes across 90 historical flights.
- ✓ **Wikipedia Web Scraping:** Extracted supplementary Falcon 9 launch logs using BeautifulSoup to parse HTML tables.
- ✓ **Data Cleaning & Feature Engineering:** Replaced missing payload values with the dataset mean (6,123 kg) and created 83 one-hot encoded features.
- ✓ **Target Variable Formulation:** Processed landing outcomes into a binary class (1 = Landed Successfully, 0 = Unsuccessful/No Attempt).

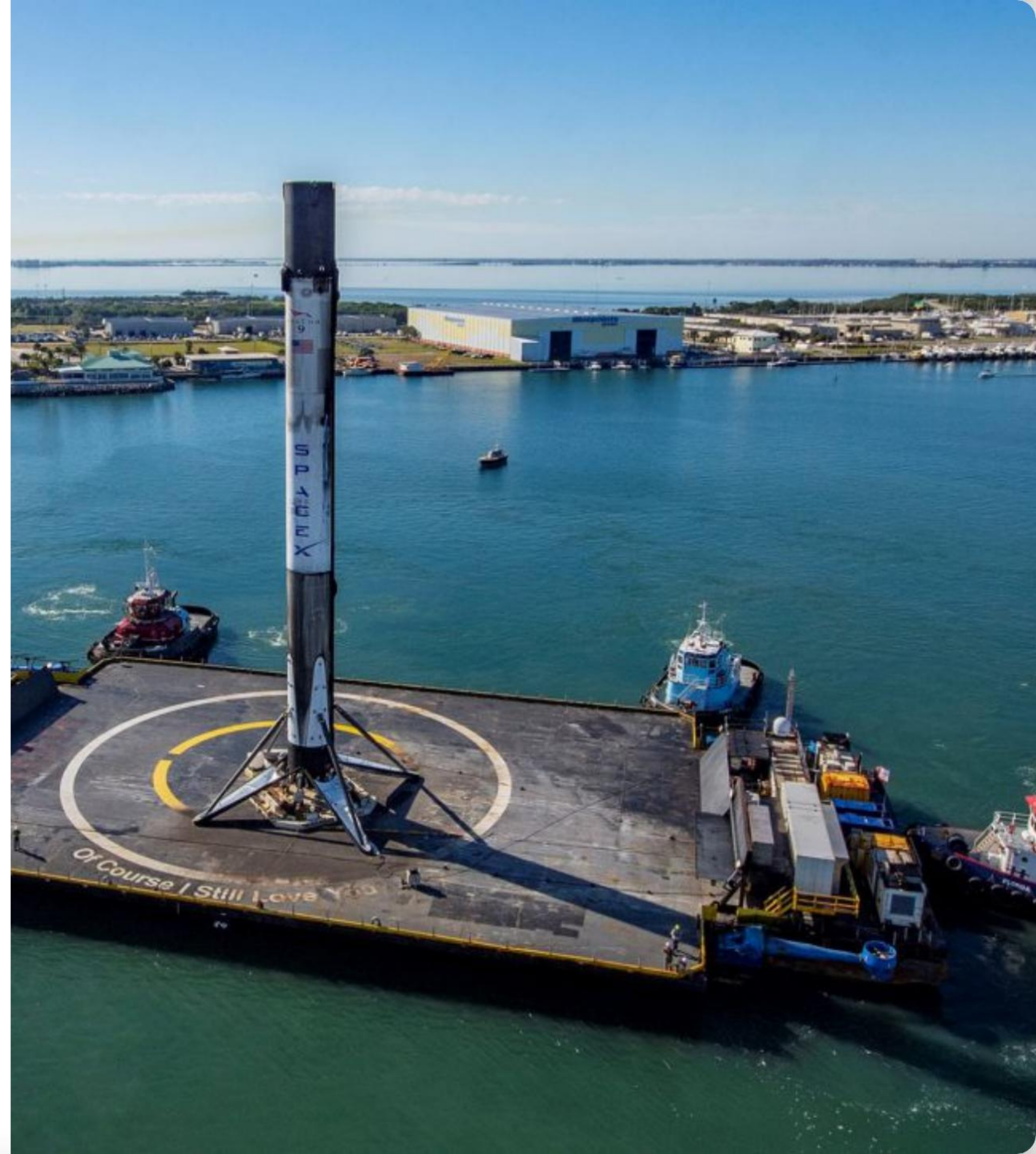


| Exploratory Data Analysis

Payload & Orbit Relationships

Landing success improved dramatically after 2017, reaching nearly 90% accuracy as SpaceX refined booster recovery telemetry.

Lower to medium payloads (<5,000 kg) demonstrated superior landing success. High-altitude orbits like ES-L1, GEO, HEO, and SSO achieved 100% booster recovery rates.



| SQL Analytics & Database EDA



Launch Site Dominance

SQL queries revealed CCAFS SLC-40 hosted over 58% of total missions, serving as the core workhorse for early commercial payloads.



NASA CRS Mass Metrics

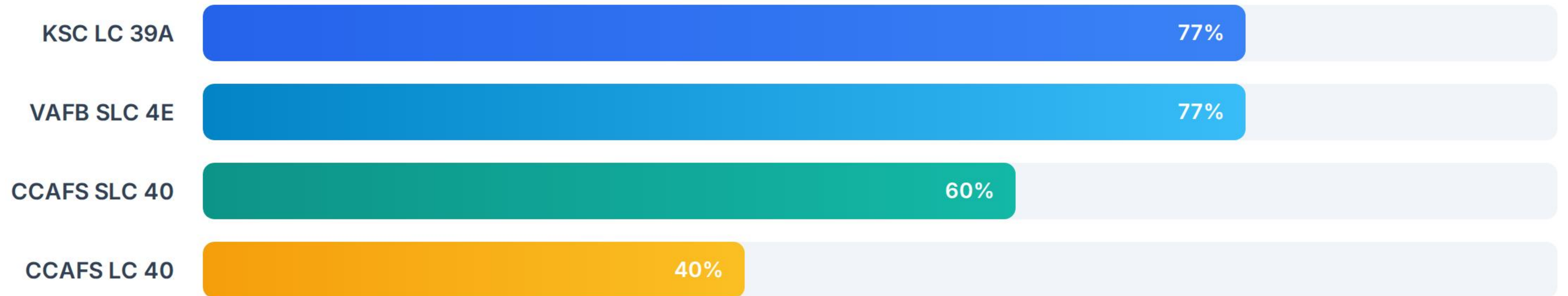
Total payload carried for NASA Resupply missions exceeded 45,000 kg with an average mission mass of 3,556 kg per flight.



Recovery Milestones

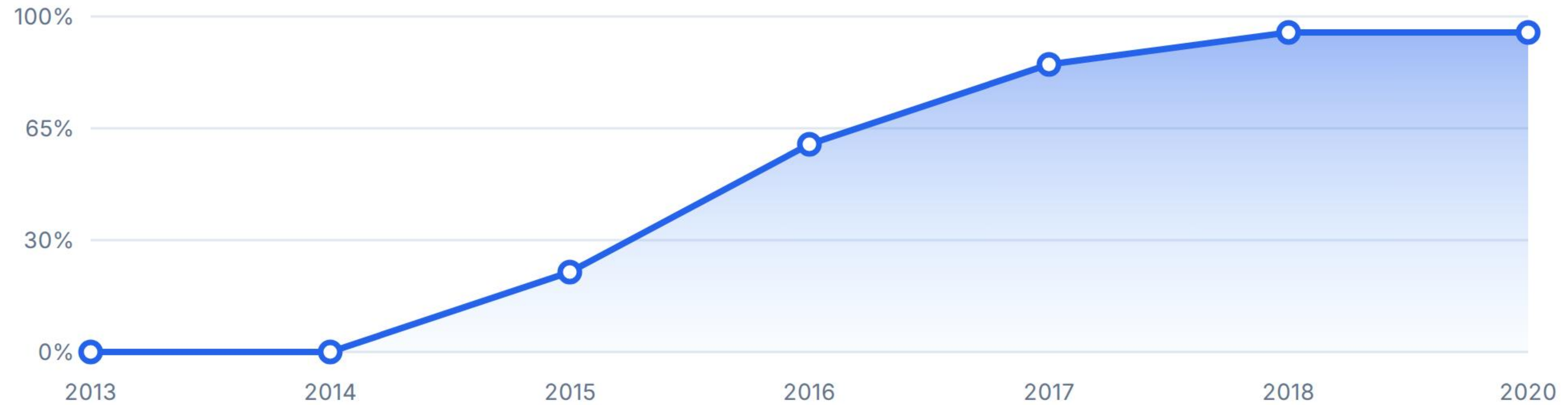
First ground pad landing success was achieved on Dec 22, 2015. Drone ship landings achieved full stability post-March 2017.

Success Rate by Launch Site



Kennedy Space Center (KSC LC 39A) and Vandenberg Air Force Base (VAFB SLC 4E) lead with a 77% success rate, benefiting from modernized pad equipment and favorable trajectory corridors.

Landing Success Rate Trend



Continuous technical iterations on grid fins, cold gas thrusters, and autonomous drone ship navigation drove recovery success from 0% in 2013 to over 88% by 2020.

| Cost Advantage & Business ROI

\$62M

SpaceX Launch Price

Disruptive Market Pricing

Traditional rocket launch providers charge upwards of **\$165 million** per launch due to expendable hardware.

By successfully landing and re-using the Falcon 9 first stage booster, SpaceX cuts operational costs by over 60%. Predicting landing success enables competing launch startups to price bids competitively while maintaining healthy profit margins.

Machine Learning Model Evaluation

Classification Algorithm	Hyperparameter Grid Tuning	Train Accuracy	Test Accuracy	Status
Logistic Regression	C=0.01, penalty='l2', solver='lbfgs'	84.7%	83.3%	Top Model
Support Vector Machine (SVM)	C=1.0, kernel='sigmoid', gamma='auto'	84.7%	83.3%	Top Model
Decision Tree Classifier	criterion='gini', splitter='best', max_depth=4	87.5%	83.3%	Top Model
K-Nearest Neighbors (KNN)	n_neighbors=10, algorithm='auto', p=1	84.7%	83.3%	Top Model

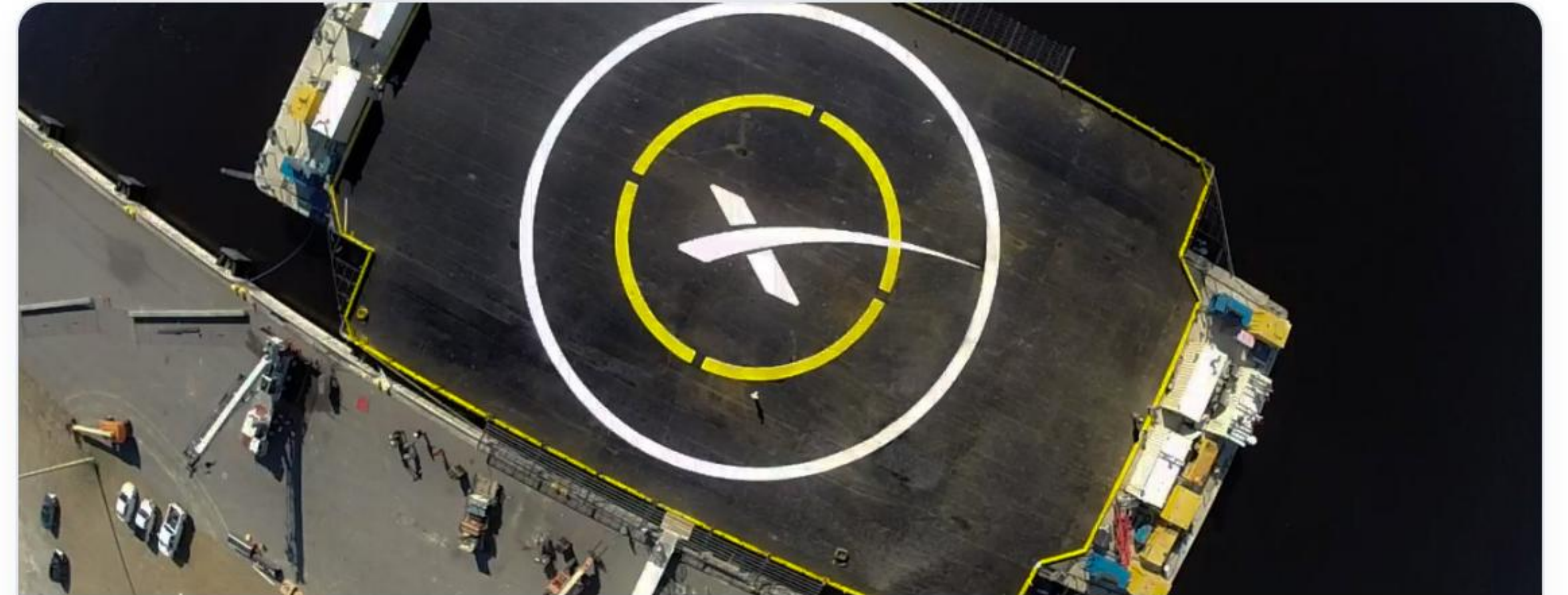
All models achieved an identical test accuracy of 83.33% across 10-fold cross-validation. Confusion matrices confirm strong precision in classifying successful landings.

Geospatial & Dashboard Analytics



Folium Interactive Map

Calculated proximity markers confirming launch pads maintain safety buffers (>14 km from cities) while situated <1 km from railways, highways, and coastlines.



Plotly Dash Application

Built dynamic dropdown filters and RangeSliders to inspect success rates across specific payload ranges (2,000 kg - 8,000 kg) and launch sites in real-time.

| Strategic Recommendations

-  **Competitive Bidding Strategy:** Target launch contracts in the \$80M–\$100M window, offering commercial clients 40% savings compared to traditional operators while preserving high gross margins.
-  **Pad Allocation Optimization:** Priority schedule high-value payloads at KSC LC 39A and VAFB SLC 4E, which demonstrate proven 77%+ booster recovery rates.
-  **Model Integration:** Deploy the trained Logistic Regression / SVM classifier into pre-flight bidding tools to dynamically assess financial risk based on payload weight and target orbit.

Questions & Discussion

Thank you for your attention!

 IBM Data Science Capstone Project | SpaceX Falcon 9 Landing Prediction

| Image Sources



http://spaceflightnow.com/wp-content/uploads/2026/01/20260114_Starlink_6-98_liftoff_AB-1.jpeg

Source: spaceflightnow.com



<https://space-offshore.com/wp-content/uploads/2020/05/ocisly-1024x724.jpg>

Source: space-offshore.com



<https://preview.redd.it/current-satellite-photo-of-39as-starship-florida-pad-v0-erlhife9tmef1.jpeg?auto=webp&s=57f5517a71a4fd56af6cc69065a09d05678038a5>

Source: www.reddit.com



http://spaceflightnow.com/wp-content/uploads/2014/12/asds_exposure_correction_no_water_smooth.jpg

Source: spaceflightnow.com